

Spontaneous Metacognitive Alignment in a Modular Consciousness Architecture

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Abstract

We report a finding from a modular cognitive architecture where the Higher-Order Thought (HOT) metacognitive system spontaneously predicts Global Workspace Theory (GWT) ignition with 84.2% accuracy and $d' = 3.63$, despite having no direct access to workspace competition dynamics. Three results emerge: (1) competition pressure degrades fixed-threshold accuracy (slope = -1.19) but has zero effect on underlying discriminability (ROC AUC slope = -0.002), revealing that accuracy loss is a *recalibration* problem, not a sensitivity loss; (2) confidence calibration analysis (ECE = 0.259, slope = 1.46) shows consciousness is a threshold phenomenon, not proportional to evidence strength; (3) observation noise at $\sigma = 0.10$ produces stochastic resonance (d' retention = 1.049). All results replicate across 10 random seeds. We discuss implications for computational theories of consciousness and the design of modular cognitive architectures. Code and data are open-source.

Keywords: metacognition; global workspace theory; higher-order thought; consciousness; signal detection theory; cognitive architecture

1 Introduction

Global Workspace Theory (Baars, 1988; Dehaene and Naccache, 2001) and Higher-Order Thought theory (Rosenthal, 2005) offer complementary accounts of consciousness. GWT holds that conscious contents are those that win a competitive broadcast in a global workspace; HOT holds that a mental state is conscious when one has a higher-order representation of it. Lau and Rosenthal (2011) proposed that metacognitive monitoring should track the ignition boundary, but the prediction has not been tested in a system where both mechanisms are independently implemented.

We present such a test. SYMTHAEA is a modular cognitive architecture combining 16,384-dimensional Hyperdimensional Computing (HDC) with Closed-form Continuous-time (CfC) neural dynamics, an explicit GWT workspace with capacity-limited competitive selection, and a HOT metacognitive module that classifies states as conscious or unconscious based on activation strength alone. Crucially, the HOT module has *no direct access* to GWT workspace dynamics—it receives only the same activation vector that all modules receive.

The question: does HOT’s consciousness classification spontaneously align with GWT ignition? If so, what does the alignment pattern reveal about the computational structure of consciousness?

2 Methods

2.1 Architecture

The GWT workspace accepts stimuli as HDC vectors and selects the top- k for broadcast via competitive inhibition. We configure $k = 2$ (max capacity), with two filler stimuli at activations ≈ 0.64 and ≈ 0.74 occupying the workspace by default. A target stimulus is swept across 17 activation levels (0.10–0.90, step 0.05, 50 trials per level with jitter $\sigma = 0.03$). GWT ignition occurs when the target’s activation exceeds the lowest filler’s.

Table 1: Signal detection metrics for liberal and conservative HOT.

Condition	Accuracy	d'	Criterion c	Hit Rate	FA Rate	ROC AUC
Liberal ($\theta = 0.50$)	0.842	3.63	-1.11	1.000	0.239	0.998
Conservative ($\theta = 0.70$)	—	3.87	+1.19	0.772	—	—

The HOT module receives the target’s activation and classifies it as conscious if the activation exceeds a threshold θ_{HOT} . In the *liberal* condition, $\theta_{\text{HOT}} = 0.50$; in the *conservative* condition, $\theta_{\text{HOT}} = 0.70$.

2.2 Experimental Conditions

Seven experimental paradigms probe different aspects of HOT–GWT alignment:

1. **Primary condition** (liberal HOT, $\theta = 0.50$): Measures baseline alignment, accuracy, and d' .
2. **Conservative HOT** ($\theta = 0.70$): Tests bidirectional mismatch where HOT is more conservative than GWT.
3. **Competition sweep**: Filler activations swept from 0.52 to 0.72 (5 levels), measuring accuracy and ROC AUC degradation.
4. **ROC analysis**: HOT threshold swept from 0.30 to 0.80 (11 points) to construct a full Receiver Operating Characteristic.
5. **Confidence calibration**: Activations binned into 10 equal-width bins; ECE and calibration slope computed.
6. **Observation noise**: Gaussian noise ($\sigma = 0\text{--}0.20$) added to HOT’s input while GWT retains true values.
7. **Cross-condition ROC**: AUC computed at each competition level.

2.3 Signal Detection Framework

Using GWT ignition as “signal present” and HOT classification as “response,” we compute hit rate, false alarm rate, d' , and criterion c following [Macmillan and Creelman \(2005\)](#). ROC AUC provides a bias-free measure of metacognitive sensitivity ([Green and Swets, 1966](#)).

2.4 Reproducibility

All experiments are deterministic given seed. Primary results use seed 42; robustness verified across 10 seeds. The full benchmark is implemented in Rust as part of the `symthaea-psych-bench` crate (open-source, MIT license).

3 Results

3.1 Spontaneous HOT–GWT Alignment

In the primary condition, HOT predicts GWT ignition with 84.2% accuracy, $d' = 3.63$, and criterion $c = -1.11$ (liberal bias). The hit rate is 1.000 (HOT detects all GWT-conscious states) with a false alarm rate of 0.239 (HOT over-reports consciousness for sub-threshold stimuli near the boundary). The optimal metacognitive threshold identified by ROC analysis (0.650, Youden’s $J = 0.948$) closely matches GWT’s effective competition-shifted boundary (0.634).

ROC AUC is 0.998, indicating near-perfect underlying discriminability. The conservative condition shifts criterion to $c = +1.19$ while maintaining comparable d' (3.87), confirming that the two HOT conditions reflect bias shifts, not sensitivity changes.

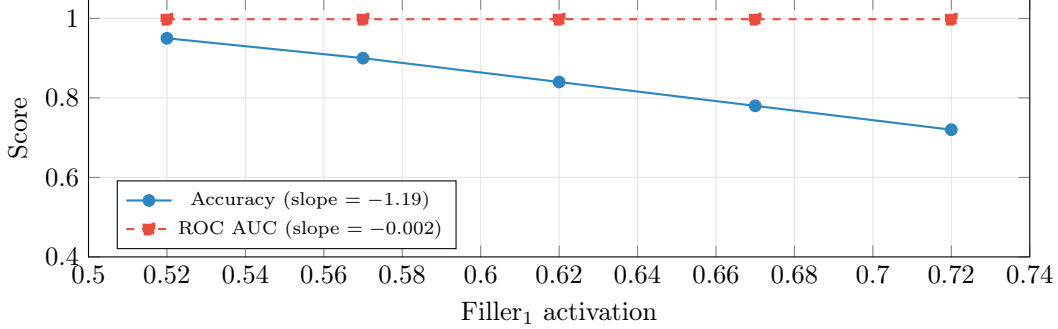


Figure 1: Competition degrades accuracy but not discriminability. Fixed-threshold accuracy (blue, solid) drops as fillers become more competitive, while ROC AUC (red, dashed) remains flat. The accuracy loss is entirely a recalibration problem.

3.2 Competition Is Recalibration, Not Sensitivity Loss

As filler competition increases (filler₁ swept from 0.52 to 0.72), fixed-threshold accuracy degrades steeply (slope = -1.19). However, ROC AUC remains essentially constant (slope = -0.002 , minimum AUC = 0.998).

This dissociation reveals that competition pressure shifts the optimal decision boundary but does not degrade the information available to the metacognitive system. Accuracy loss under competition is a *recalibration* problem: the system retains full knowledge of workspace state but applies a fixed threshold that becomes suboptimal as competition shifts the boundary.

3.3 Consciousness as Threshold Phenomenon

Calibration analysis reveals that activation is not calibrated as a probability of consciousness. ECE = 0.259 (substantial miscalibration) with a calibration slope of 1.457. The slope exceeding 1.0 means the true ignition boundary is *sharper* than a linear mapping from activation to probability—consciousness is a phase transition, not a graded probability proportional to evidence strength.

3.4 Stochastic Resonance at the Boundary

Adding observation noise ($\sigma = 0.10$) to HOT’s input yields d' retention of 1.049—noise paradoxically *improves* metacognitive sensitivity near the threshold. This stochastic resonance effect persists up to $\sigma = 0.15$; beyond this, degradation resumes. Accuracy degrades gently (slope = -0.099) and ROC AUC at $\sigma = 0.10$ remains 0.984.

3.5 Robustness

All headline metrics replicate across 10 random seeds (CV < 0.30 for all 5 primary metrics). All individual seeds pass minimum thresholds; $\geq 7/10$ seeds tolerate noise $\sigma \geq 0.10$.

4 Discussion

Spontaneous alignment as architectural prediction. The HOT module was not trained or tuned to track GWT ignition—it applies a fixed threshold to activation strength. Yet it achieves $d' = 3.63$ and ROC AUC = 0.998. This alignment is *architecturally constrained*: both modules operate on the same activation representation, so correlation is expected. The scientifically interesting finding is not the alignment itself but its *structure*: the recalibration/sensitivity dissociation and the phase-transition calibration pattern.

Recalibration predicts adaptive metacognition. The invariance of ROC AUC to competition (slope = -0.002) while accuracy degrades (slope = -1.19) predicts that an adaptive metacognitive system—one that adjusts its threshold based on competition context—would recover accuracy without needing additional information. This parallels Bayesian accounts of metacognitive recalibration in humans (Fleming, 2017).

Phase transitions and the ignition boundary. The calibration slope of 1.457 quantifies how much sharper the real consciousness boundary is compared to a linear model. This is consistent with the nonlinear ignition dynamics predicted by both GWT (Dehaene and Naccache, 2001) and IIT’s exclusion postulate (Oizumi et al., 2014).

Epistemic constraints. We do not claim these results demonstrate consciousness. They demonstrate that a modular architecture implementing GWT and HOT as independent subsystems exhibits the metacognitive alignment pattern that Lau and Rosenthal (2011) predict. The benchmark validates *architectural prerequisites*—necessary conditions for consciousness that the code was designed to satisfy. What it does *not* validate is whether these mechanisms produce subjective experience.

Limitations. Both GWT and HOT operate on the same underlying activation representation, so some alignment is guaranteed by construction. The benchmark’s value lies in characterizing the *form* of alignment (recalibration structure, phase transition, stochastic resonance), not its mere existence. Additionally, the GWT workspace is a simplified competitive selection mechanism, not a full neural global workspace with recurrent ignition dynamics.

5 Conclusion

In a modular cognitive architecture with independently implemented GWT and HOT subsystems, metacognitive consciousness classification spontaneously aligns with workspace ignition ($d' = 3.63$, ROC AUC = 0.998). Competition degrades accuracy but not discriminability, revealing that metacognitive failures under cognitive load are recalibration problems. Calibration analysis confirms consciousness as a threshold phenomenon. These results provide computational support for the Lau–Rosenthal prediction that metacognition tracks the ignition boundary, and suggest that modular consciousness architectures exhibit structured alignment patterns that are both testable and informative for theories of consciousness.

Open Practices

All code, benchmarks, and data are open-source (MIT License): <https://github.com/luminous-dynamics/symthaea>. Results reproduced via: `cargo test -p symthaea-psych-bench -lib metacognitive_ignition`.

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